Syon Mansur

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EDUCATION

Ph.D., Neuroscience | University of California, Los Angeles — Sept 2025 – Present
Advisor: Joel Zylberberg

B.S., Neuroscience; Minor in Bioinformatics | University of California, Los Angeles — Sept 2021 – June 2025

RESEARCH EXPERIENCE

Graduate Researcher | Zylberberg Lab, UCLA — Jan 2026 – Present

- Showed that widening CNNs lets greedy layer-wise self-supervised training (Barlow Twins, SimCLR) match or beat end-to-end backpropagation (78.3% vs. 75.4% kNN accuracy in the widest networks), and linked the effect to representational geometry (arXiv 2610.00753).*
- Extended SAILnet, a spiking model of V1 with local plasticity, with a learning rule approximating the Barlow Twins invariance term; trained on natural images sampled along recorded human eye drift, the network learned phase-invariant complex cells (59% of units, F1/F0 < 1).
- Implemented difference target propagation and predictive coding as backprop replacements in a PyTorch deep Q-learning agent on a visual sequence-violation task.*

Research Assistant | Hollon Lab, UC San Diego — June 2024 – Sept 2024

- Developed a pipeline to detect and quantify spontaneous activity in fiber-photometry recordings.
- Identified stereotyped behaviors in mice during optogenetic stimulation using pose tracking (SLEAP, DeepLabCut).

Research Intern | Mucke Lab, Gladstone Institutes (UCSF) — June 2023 – Sept 2023

- Built an automated Snakemake pipeline (Python, R) that processes raw EEG into brain-state labels, epileptiform-event detections, and spectral features; used to process $\sim$100 24-hour recordings.*
- Performed single-cell RNA sequencing to investigate the role of tau in excitatory neurons; used immunohistochemistry to test follow-up hypotheses.

Research Assistant | Wikenheiser Lab, UCLA — Oct 2021 – June 2025

- Led a project investigating methamphetamine's effects on hippocampal place cells; built a MATLAB analysis pipeline revealing disrupted backward place-field expansion and theta phase precession.
- Designed and ran experiments on dopamine projections across brain regions to study foraging and effort-based decision making.

Workshop Member | Telluride Neuromorphic Engineering Workshop — July 2020 – Aug 2020

- Built simulations of neural circuits in Nengo (Python) and LTspice; designed experiments probing guided reinforcement learning and imitation learning.

PUBLICATIONS & PRESENTATIONS

- **Mansur, S.**, & Zylberberg, J. Increasing width allows greedy layer-wise training to rival end-to-end backpropagation in self-supervised learning. *Under review, ICLR 2027.* arXiv:2610.00753
- Hadjas, L. C., Kollman, G. J., Linderhof, L., Xia, M., **Mansur, S.**, Saint-Pierre, M., Lim, B. K., Lee, E. B., Cicchetti, F., Awatramani, R., Hollon, N. G., & Hnasko, T. S. Parkinson's disease-vulnerable and -resilient dopamine neurons display opposite responses to excitatory input. *bioRxiv; in review, Nature Communications.*
- **Mansur, S.**, & Zylberberg, J. Why are brains shallow and wide? Increased width allows local learning to rival backpropagation. *Abstract submitted, COSYNE 2027.*
- **Mansur, S.**, Mueller, L., & Wikenheiser, A. Quantifying the effect of methamphetamine on hippocampal place cells. *Poster, 25th Annual UCLA Neuroscience Poster Day and UCLA Decision Making Affinity Group Symposium, 2024.*

AWARDS

NSF Graduate Research Fellowship (GRFP) — 2026
Scheibel Scholar, UCLA Life Science Dean's Award for Neuroscience Research — 2023–24, 2024–25
UCLA Undergraduate Research Fellowship — 2024

Languages: Python, C++, MATLAB, R, Bash
Skills: PyTorch, scikit-learn, NumPy, SciPy, pandas, Matplotlib

*Code available at github.com/SyonMansur